

VEX IQ Robotics Kits

WHAT'S IN THE KIT

- 15 VEX IQ kits — brain, motors, sensors, batteries, wheels, structural parts
- Build and coding instructions
- Accompanying Cube Kits also available



QUICK-START GUIDE

1. Follow the included build instructions for the base robot ('Clawbot'-style).
2. Code with VEXcode IQ (blocks or Python) on Chromebooks.
3. Driver control works out of the box for early wins.
4. Budget 2-3 sessions for the first build — it's part of the learning.

FULL LESSONS IN THIS GUIDE

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PRO TIP: 15 kits supports a whole class in teams of 2 — the largest robotics experience in the LoT.

VEX IQ Robotics Kits

LESSON · GRADES 3-5

Build Team Robotics

LEARNING OUTCOMES

- Students will follow multi-step technical build instructions
- Students will work productively in defined team roles
- Students will operate a robot with driver control

MATERIALS

- VEX IQ kits (1 per team of 2-3)
- Build instructions (included)
- Role cards: parts manager, builder, checker
- Cooperative challenge: shared field with game objects

INSTRUCTIONS

1. Unbox with inventory: the parts manager confirms key components.
2. Build across 2-3 sessions with roles rotating every 30 minutes.
3. Checker's job: verify each step against the diagram before moving on.
4. First motion: connect the controller and drive — a huge milestone, celebrate it.
5. Cooperative challenge: teams work together to move objects to goals.
6. Reflection: which role was hardest? What did the diagrams not tell you?

ACTIVITY

Build Team Robotics — a multi-session build where following technical diagrams IS the curriculum. Rotating roles keeps every student building, checking, and managing parts.

VEX IQ Robotics Kits

LESSON · GRADES 6-8

Cube Collection Showdown

LEARNING OUTCOMES

- Students will modify a robot design for a specific game task
- Students will program driver-assist and autonomous routines
- Students will develop competitive strategy under constraints

MATERIALS

- VEX IQ kits + Cube Kits
- Devices with VEXcode IQ
- Match field with scoring zones
- Match schedule and scoreboard

INSTRUCTIONS

1. Present the game: cubes, scoring zones, match time, alternating driver/auto periods.
2. Teams analyze: what claw, pusher, or lift design fits their strategy?
3. Modify base robots for the cube task; test grabbing and releasing.
4. Program one autonomous routine for the match's auto period.
5. Scrimmage day: informal matches expose weaknesses; pit repairs encouraged.
6. Tournament: round-robin matches with alliance scoring like real VEX.

ACTIVITY

Cube Collection Showdown — modify, program, and compete in alternating driver and autonomous periods. Pit repairs between matches teach more engineering than the build did.

VEX IQ Robotics Kits

LESSON · GRADES 9-12

Autonomous Engineering Cycle

LEARNING OUTCOMES

- Students will program sensor-driven autonomous routines
- Students will maintain an engineering notebook to competition standard
- Students will iterate systematically toward performance goals

MATERIALS

- VEX IQ kits with sensors (distance, color, gyro)
- Devices with VEXcode IQ (blocks or Python)
- Engineering notebooks (physical or digital template)
- Autonomous challenge field with repeatable setup

INSTRUCTIONS

1. Define the autonomous challenge: navigate, detect, manipulate, return — no driver.
2. Notebook protocol: every session logs goal, changes, test results, next steps.
3. Teams integrate sensors: gyro turns beat timed turns — let them discover why.
4. Weekly benchmark runs with recorded scores track real progress.
5. Mid-project design review: teams present notebooks and defend decisions.
6. Final runs plus notebook assessment, mirroring real VEX judging.

ACTIVITY

Autonomous Engineering Cycle — sensor-driven autonomy with competition-standard documentation. The notebook requirement transforms tinkering into engineering practice colleges recognize.